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ELECTRONIC PACKAGE AND MANUFACTURING METHOD THEREOF

Abstract

An electronic package is provided, in which a mesh structure is disposed between a circuit structure and an electronic element to increase the shunt path of current. Therefore, when the electronic element is used as an electrode pad of a power contact, the current can be passed through a conductive sheet of the circuit structure via the mesh structure, such that the power loss can be reduced and the IR drop of the electronic element can meet the requirements.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a Continuation of U.S. patent application Ser. No. 18/056,078 filed Nov. 16, 2022, which claims benefit of priority to Taiwanese Patent Application No. 111133425 filed Sep. 2, 2022, the entire contents of which are incorporated herein by reference.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to a semiconductor packaging process, and more particularly, to an electronic package for improving reliability and a manufacturing method thereof.

2. Description of Related Art

[0003] With the vigorous development of the electronic industry, electronic products are also gradually moving towards the trend of multi-function and high performance. In order to meet the packaging requirements of miniaturization of electronic packages, the technology of embedded packaging has been developed.

[0004] FIG. 1 is a schematic cross-sectional view of a conventional semiconductor package 1. As shown in FIG. 1, a semiconductor element 12 is embedded in an encapsulation layer 13. Next, a circuit structure 14 is formed on the encapsulation layer 13 and an active surface 12a of the semiconductor element 12, so that the circuit structure 14 is electrically connected to electrode pads 120 of the semiconductor element 12. After that, an insulating protection layer 15 is formed on the circuit structure 14, and parts of the surface of the circuit structure 14 are exposed from the insulating protection layer 15 for bonding conductive elements 16 such as solder balls.

[0005] However, in the conventional semiconductor package 1, the circuit of the circuit structure 14 is made of copper. Therefore, when the circuit structure 14 is designed to be thin and the thickness of the circuit is extremely thin, after the electrode pads 120 of the semiconductor element 12 are connected to the circuit of the circuit structure 14, the electrode pad 120 serving as a power source will suffer a great loss, so that its IR drop (e.g., voltage drop) does not meet the requirements.

[0006] Therefore, how to overcome the above-mentioned drawbacks of the prior art has become an urgent issue to be solved at present.

SUMMARY

[0007] In view of the various deficiencies of the prior art, the present disclosure provides an electronic package, comprising: a circuit structure defined with a first side and a second side opposing the first side, wherein the circuit structure is provided with a conductive sheet and a circuit layer spaced apart from the conductive sheet on the first side of the circuit structure; an electronic element disposed on the first side of the circuit structure, wherein the electronic element has a plurality of electrode pads, and at least one of the plurality of electrode pads is electrically connected to the circuit layer; and a mesh structure disposed between the electronic element and

the first side of the circuit structure, wherein the mesh structure is electrically connected to the conductive sheet and at least the other one of the plurality of electrode pads, wherein the mesh structure has a plurality of interlaced strips to form a plurality of mesh holes corresponding to the plurality of electrode pads.

[0008] The present disclosure also provides a method of manufacturing an electronic package, the method comprising: providing a circuit structure, wherein the circuit structure is defined with a first side and a second side opposing the first side, and the circuit structure is provided with a conductive sheet and a circuit layer spaced apart from the conductive sheet on the first side of the circuit structure; disposing an electronic element having a plurality of electrode pads on the first side of the circuit structure, wherein at least one of the plurality of electrode pads is electrically connected to the circuit layer; and disposing a mesh structure between the electronic element and the first side of the circuit structure, wherein the mesh structure is electrically connected to the conductive sheet and at least the other one of the plurality of electrode pads, wherein the mesh structure has a plurality of interlaced strips to form a plurality of mesh holes corresponding to the plurality of electrode pads.

[0009] In the aforementioned electronic package and method, at least one of the plurality of electrode pads is used as a signal contact, and at least the other one of the plurality of electrode pads is used as a power contact.

[0010] In the aforementioned electronic package and method, the present disclosure further comprises forming a plurality of conductive bumps on the plurality of electrode pads, wherein a part of the plurality of conductive bumps is electrically connected to the circuit layer, and another part of the plurality of conductive bumps is electrically connected to the conductive sheet. The present disclosure also comprises providing the mesh structure and a protective layer on the electronic element, wherein the protective layer covers the mesh structure, the plurality of electrode pads and the plurality of conductive bumps. For example, the mesh structure and each of the plurality of conductive bumps are coplanar. Alternatively, a width of each of the strips is greater than a maximum diameter of an end surface of each of the plurality of conductive bumps.

[0011] In the aforementioned electronic package and method, a width of each of the strips is at least 10 micrometers.

[0012] In the aforementioned electronic package and method, the mesh structure is a metal mesh.

[0013] In the aforementioned electronic package and method, the present disclosure further comprises disposing the mesh structure on the electronic element; covering the electronic element and the mesh structure by an encapsulation layer, wherein the electronic element is embedded in the encapsulation layer; and forming the circuit structure on the encapsulation layer.

[0014] In the aforementioned electronic package and method, the present disclosure further comprises disposing the mesh structure on the circuit structure; disposing the electronic element on the circuit structure and the mesh structure in a flip-chip manner; and forming an encapsulation layer between the electronic element and the circuit structure, wherein the encapsulation layer covers the mesh structure.

[0015] As can be seen from the above, in the electronic package and the manufacturing method thereof of the present disclosure, the mesh structure is disposed between the electronic element and the circuit structure, so as to increase the shunt path of current. Therefore, compared with the prior art, in the electronic package of the present disclosure, the current at the electrode pads serving as power contacts can be passed through the conductive sheet to the second side of the circuit structure via a conductive area composed of the mesh structure and the conductive bumps, such that the power loss can be reduced, so that the IR drop of the electronic element can meet the requirements.

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] FIG. 1 is a schematic cross-sectional view of a conventional semiconductor package.

[0017] FIG. 2A to FIG. 2D are schematic cross-sectional views illustrating a method of manufacturing an electronic package according to a first embodiment of the present disclosure.

[0018] FIG. 3A to FIG. 3B are schematic cross-sectional views illustrating a method of manufacturing an electronic package according to a second embodiment of the present disclosure.

[0019] FIG. 4 is a schematic partial top view of the electronic package according to the present disclosure.

DETAILED DESCRIPTIONS

[0020] The following describes the implementation of the present disclosure with examples. Those skilled in the art can easily understand other advantages and effects of the present disclosure from the contents disclosed in this specification.

[0021] It should be understood that, the structures, ratios, sizes, and the like in the accompanying figures are used for illustrative purposes to facilitate the perusal and comprehension of the contents disclosed in the present specification by one skilled in the art, rather than to limit the conditions for practicing the present disclosure. Any modification of the structures, alteration of the ratio relationships, or adjustment of the sizes without affecting the possible effects and achievable proposes should still be deemed as falling within the scope defined by the technical contents disclosed in the present specification. Meanwhile, terms such as “on,” “upper,” “first,” “second,” “one,” “a,” and the like used herein are merely used for clear explanation rather than limiting the practicable scope of the present disclosure, and thus, alterations or adjustments of the relative relationships thereof without essentially altering the technical contents should still be considered in the practicable scope of the present disclosure.

[0022] FIG. 2A to FIG. 2D are schematic cross-sectional views illustrating a method of manufacturing an electronic package 2 according to a first embodiment of the present disclosure. In an embodiment, the electronic package 2 adopts the wafer level packaging specification.

[0023] As shown in FIG. 2A, a carrier 20 having a release layer 200 is provided, and an electronic element 22 having a mesh structure 21 is disposed on the release layer 200.

[0024] In an embodiment, the carrier 20 is, for example, a plate of semiconductor material (e.g., silicon or glass), on which the release layer 200 is formed.

[0025] Furthermore, the electronic element 22 is an active element, a passive element, or a combination of the active element and the passive element, where the active element is, for example, a semiconductor chip, and the passive element is, for example, a resistor, a capacitor, or an inductor. For example, the electronic element 22 is an active element and has an active surface 22a and an inactive surface 22b opposing the active surface 22a, and the electronic element 22 is disposed on the release layer 200 with the inactive surface 22b thereof, wherein the active surface 22a has a plurality of electrode pads 220, 220a for conductive bumps 221, 221a such as copper pillars or solder balls to be formed on the electrode pads 220, 220a, and the mesh structure 21 and a protective layer 222 are formed on the active surface 22a, so that the protective layer 222 covers the mesh structure 21, the electrode pads 220, 220a and the conductive bumps 221, 221a, wherein some of the electrode pads 220 are used as power contacts, and some of the electrode pads 220a are used as signal contacts.

[0026] Further, the conductive bumps 221, 221a and the mesh structure 21 may be exposed from the protective layer 222. For example, the protective layer 222 is a non-conductive film (NCF), and a part of the protective layer 222 can be removed by a leveling process, such as etching or grinding, so that end surfaces of the conductive bumps 221, 221a and an upper surface of the mesh structure 21 are flush with an upper surface of the protective layer 222.

[0027] In addition, the mesh structure 21 is a metal mesh, which has a plurality of interlaced strips 21a to form mesh holes 210 (e.g., curved grooves as shown in FIG. 4) corresponding to the

electrode pads **220a**, so that the conductive bumps **221a** connected to the signal contacts (the electrode pads **220a**) are located in the mesh holes **210**, and the conductive bumps **221** connected to the power contacts (the electrode pads **220**) are connected to the strips **21a** of the mesh structure **21** (as shown in FIG. 4), such that the protective layer **222** fills the remaining space of the mesh holes **210**. For example, the mesh structure **21** and the conductive bumps **221**, **221a** can be simultaneously fabricated via an electroplating process, so that the mesh structure **21** and the conductive bumps **221**, **221a** have the same height, that is, they are coplanar. However, in other embodiments, the conductive bumps **221**, **221a** can also be higher than the mesh structure **21**, so that the conductive bumps **221**, **221a** protrude from the mesh structure **21**.

[0028] In addition, a width d of each of the strips **21a** of the mesh structure **21** is greater than a maximum diameter r of the end surface of each of the conductive bumps **221**, **221a**, as shown in FIG. 4. For example, the width d of each of the strips **21a** of the mesh structure **21** is at least 10 micrometers (μm).

[0029] As shown in FIG. 2B, an encapsulation layer **23** is formed on the carrier **20**, so that the encapsulation layer **23** covers the electronic element **22** and the protective layer **222**.

[0030] In an embodiment, the material for forming the encapsulation layer **23** is an insulating material, such as polyimide (PI), dry film, encapsulation colloid such as epoxy resin, or a molding compound, and the encapsulation layer **23** can be formed on the carrier **20** by lamination or molding.

[0031] Furthermore, a leveling process, such as grinding, can remove a part of the encapsulation layer **23**, and even remove a part of the protective layer **222** and/or a part of the conductive bumps **221**, **221a**, so that an upper surface of the encapsulation layer **23** is flush with the upper surface of the protective layer **222**, such that the conductive bumps **221**, **221a** on the electronic element **22** are exposed from the encapsulation layer **23**.

[0032] As shown in FIG. 2C, a redistribution layer (RDL) process is performed to form a circuit structure **24** on the encapsulation layer **23** and the electronic element **22**, and the circuit structure **24** is electrically connected to the electronic element **22** and the mesh structure **21**.

[0033] In an embodiment, the circuit structure **24** includes at least one dielectric layer **240**, and a circuit layer **241** disposed on the dielectric layer **240** and electrically connected to the conductive bumps **221a**, wherein the outermost circuit layer **241** has electrical contact pads **242** for bonding conductive elements **25** such as solder material. For example, the material for forming the circuit layer **241** is copper, and the material for forming the dielectric layer **240** is, for example, Ajinomoto build-up film (ABF), polybenzoxazole (PBO), polyimide (PI), prepreg (PP), or other dielectric materials, or a solder-proof material such as solder resist and graphite.

[0034] Furthermore, the circuit structure **24** is defined with a first side **24a** and a second side **24b** opposing the first side **24a**, and the circuit structure **24** is bonded onto the encapsulation layer **23** with the first side **24a** thereof, so that the conductive elements **25** are disposed on the electrical contact pads **242** of the second side **24b**. For example, the circuit structure **24** has a conductive sheet **243** contacting the conductive bumps **221** and the mesh structure **21** on the first side **24a** thereof, and the conductive sheet **243** is spaced apart from the circuit layer **241** of the first side **24a**, so that the conductive sheet **243** is electrically connected to the mesh structure **21** and the conductive bumps **221** on the electrode pads **220** serving as power contacts, and the circuit layer **241** of the first side **24a** is electrically connected to the conductive bumps **221a** on the electrode pads **220a** serving as signal contacts.

[0035] As shown in FIG. 2D, the carrier **20** and the release layer **200** are removed to expose the encapsulation layer **23** and the inactive surface **22b** of the electronic element **22** to obtain the electronic package **2** of the present disclosure.

[0036] In an embodiment, the exposed surface of the encapsulation layer **23** is flush with the inactive surface **22b** of the electronic element **22**, so that the inactive surface **22b** of the electronic element **22** is exposed from the encapsulation layer **23** for the electronic element **22** to dissipate

heat.

[0037] Furthermore, the electronic package **2** can be connected to at least one electronic device such as a package module, a circuit board, a chip, or other electronic devices via the conductive elements **25**.

[0038] Therefore, in the manufacturing method of the present disclosure, the mesh structure **21** is designed to increase the shunt path of the current. Therefore, compared with the prior art, even though the circuit structure **24** is designed to be thin and the thicknesses of the circuit layer **241** and the conductive sheet **243** are extremely thin, after the electronic element **22** is bonded to the circuit structure **24**, the current at the electrode pads **220** serving as power contacts can be passed through the conductive sheet **243** to the conductive elements **25** of the second side **24b** via a conductive area composed of the mesh structure **21** and the conductive bumps **221**, such that the power loss can be reduced, so that the IR drop (e.g., voltage drop) of the electronic element **21** can meet the requirements.

[0039] Furthermore, the larger the layout area of the mesh structure **21** (or the thicker the width *d* of the strip **21a**), the better the effect of reducing power loss. For example, when the width *d* of the strip **21a** is greater than or equal to 10 micrometers (μm), the IR drop of the electronic element **21** can be reduced from 19.41 (without the mesh structure **21**) to 14.09, which is about 25% lower.

[0040] FIG. 3A to FIG. 3B are schematic cross-sectional views illustrating a method of manufacturing an electronic package **3** according to a second embodiment of the present disclosure. The difference between the second embodiment and the first embodiment lies in the packaging form, and other designs are substantially the same, so the similarities will not be repeated below. In an embodiment, the electronic package **3** adopts the flip-chip package specification.

[0041] As shown in FIG. 3A, the mesh structure **21** and the plurality of conductive bumps **221**, **221a** are formed on the first side **24a** of the circuit structure **24**.

[0042] In an embodiment, the circuit structure **24** is in the form of a carrier board, such as a package substrate with a core layer or a coreless package substrate. However, there are various types of the circuit structure **24**, as long as it is a carrier board, such as a silicon interposer, for carrying electronic elements such as chips, and the present disclosure is not limited to as such.

[0043] Furthermore, the mesh structure **21** and the conductive bumps **221**, **221a** can be simultaneously fabricated on the surface of the first side **24a** of the circuit structure **24** by an electroplating process, and the first side **24a** of the circuit structure **24** is provided with the conductive sheet **243**.

[0044] As shown in FIG. 3B, the electronic element **22** is bonded onto the conductive bumps **221**, **221a** with the electrode pads **220**, **220a** thereof in a flip-chip manner, so that the mesh structure **21** is located between the electronic element **22** and the first side **24a** of the circuit structure **24**.

Afterwards, an encapsulation layer **33** such as an underfill is formed between the electronic element **22** and the first side **24a** of the circuit structure **24**, so that the encapsulation layer **33** covers the mesh structure **21** and the conductive bumps **221**, **221a**.

[0045] In an embodiment, the conductive elements **25** may be disposed on the second side **24b** of the circuit structure **24**.

[0046] Therefore, in the manufacturing method of the present disclosure, the mesh structure **21** is designed to increase the shunt path of the current. Therefore, compared with the prior art, after the electronic element **22** is bonded to the circuit structure **24**, the current at the electrode pads **220** serving as power contacts can be passed through the conductive sheet **243** to the conductive elements **25** of the second side **24b** via a conductive area composed of the mesh structure **21** and the conductive bumps **221**, such that the power loss can be reduced, so that the IR drop of the electronic element **22** can meet the requirements.

[0047] The present disclosure further provides an electronic package **2**, **3**, which comprises: a circuit structure **24**, at least one electronic element **22**, and a mesh structure **21**.

[0048] The circuit structure **24** is defined with a first side **24a** and a second side **24b** opposing the

first side **24a**, wherein the circuit structure **24** is provided with a conductive sheet **243** and a circuit layer **241** spaced apart from the conductive sheet **243** on the first side **24a** of the circuit structure **24**.

[0049] The electronic element **22** is disposed on the first side **24a** of the circuit structure **24**, wherein the electronic element **22** has a plurality of electrode pads **220**, **220a**, and at least one of the plurality of electrode pads **220a** is electrically connected to the circuit layer **241**.

[0050] The mesh structure **21** is disposed between the electronic element **22** and the first side **24a** of the circuit structure **24** to electrically connect the conductive sheet **243** and at least the other one of the plurality of electrode pads **220**, wherein the mesh structure **21** has a plurality of interlaced strips **21a** to form mesh holes **210** at corresponding portions of the electrode pads **220a**.

[0051] In one embodiment, at least one of the plurality of electrode pads **220a** is used as a signal contact, and at least another of the plurality of electrode pads **220** is used as a power contact.

[0052] In one embodiment, conductive bumps **221**, **221a** are formed on the plurality of electrode pads **220**, **220a**, so that a part of the conductive bumps **221a** is electrically connected to the circuit layer **241**, and another part of the conductive bumps **221** is electrically connected to the conductive sheet **243**. Further, the mesh structure **21** and a protective layer **222** are formed on the electronic element **22**, so that the protective layer **222** covers the mesh structure **21**, the plurality of electrode pads **220**, **220a** and the plurality of conductive bumps **221**, **221a**. For example, the mesh structure **21** is coplanar with the conductive bumps **221**, **221a**. Alternatively, a width d of each of the strips **21a** is greater than a maximum diameter r of an end surface of each of the conductive bumps **221**, **221a**.

[0053] In one embodiment, the width d of each of the strips **21a** is at least 10 micrometers.

[0054] In one embodiment, the mesh structure **21** is a metal mesh.

[0055] In one embodiment, the electronic package **2** further includes an encapsulation layer **23** bonded to the circuit structure **24**, such that the electronic element **22** is embedded in the encapsulation layer **23**.

[0056] In one embodiment, the electronic package **2** further includes an encapsulation layer **33** formed between the electronic element **22** and the circuit structure **24** to cover the mesh structure **21**.

[0057] To sum up, in the electronic package and the manufacturing method thereof of the present disclosure, the mesh structure is designed to increase the shunt path of the current. Therefore, in the electronic package of the present disclosure, the current of the electrode pads serving as power contacts can be passed through the conductive sheet to the second side of the circuit structure via the mesh structure and the conductive bumps, such that the power loss can be reduced, so that the IR drop of the electronic element can meet the requirements.

[0058] The foregoing embodiments are provided for the purpose of illustrating the principles and effects of the present disclosure, rather than limiting the present disclosure. Anyone skilled in the art can modify and alter the above embodiments without departing from the spirit and scope of the present disclosure. Therefore, the scope of protection with regard to the present disclosure should be as defined in the accompanying claims listed below.

Claims

1. An electronic package, comprising: a circuit structure defined with a first side and a second side opposing the first side, wherein the circuit structure is provided with a conductive sheet and a circuit layer spaced apart from the conductive sheet on the first side of the circuit structure; an electronic element disposed on the first side of the circuit structure, wherein the electronic element has a plurality of electrode pads, and at least one of the plurality of electrode pads is electrically connected to the circuit layer; and a mesh structure disposed between the electronic element and the first side of the circuit structure, wherein the mesh structure is electrically connected to the

conductive sheet and at least the other one of the plurality of electrode pads.

2. The electronic package of claim 1, wherein at least one of the plurality of electrode pads is used as a signal contact, and at least the other one of the plurality of electrode pads is used as a power contact.

3. The electronic package of claim 1, further comprising a plurality of conductive bumps formed on the plurality of electrode pads, wherein a part of the plurality of conductive bumps is electrically connected to the circuit layer, and another part of the plurality of conductive bumps is electrically connected to the conductive sheet.

4. The electronic package of claim 3, wherein the mesh structure has a plurality of interlaced strips to form a plurality of mesh holes corresponding to the plurality of electrode pads.

5. The electronic package of claim 3, wherein the electronic element is provided with the mesh structure and a protective layer thereon, and wherein the protective layer covers the mesh structure, the plurality of electrode pads and the plurality of conductive bumps.

6. The electronic package of claim 3, wherein the mesh structure and each of the plurality of conductive bumps are coplanar.

7. The electronic package of claim 4, wherein a width of each of the strips is greater than a maximum diameter of an end surface of each of the plurality of conductive bumps.

8. The electronic package of claim 4, wherein a width of each of the strips is at least 10 micrometers.

9. The electronic package of claim 1, wherein the mesh structure is a metal mesh.

10. The electronic package of claim 1, further comprising an encapsulation layer bonded to the circuit structure, wherein the electronic element is embedded in the encapsulation layer.

11. The electronic package of claim 1, further comprising an encapsulation layer formed between the electronic element and the circuit structure to cover the mesh structure.
